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$$\begin{aligned}& \int \frac{\sin x + x \cos x}{x \sin x} dx \\&= \int \frac{\sin x}{x \sin x} dx + \int \frac{x \cos x}{x \sin x} dx = \int \frac{1}{x} dx + \int \frac{\cos x}{\sin x} dx \\&= \ln|x| + c + \int \frac{\cos x}{\sin x} dx \\&u = \sin x \Rightarrow du = \cos x dx \\&\Rightarrow \ln|x| + c + \int \frac{1}{u} du \\&= \ln|x| + \ln|\sin x| + c\end{aligned}$$

References